

Julian R. Grossman

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EDUCATION

Pennsylvania State University - B.S. Computational Data Science | Expected May 2027 | Cumulative GPA: 3.0

Relevant Coursework: Artificial Intelligence, Data Structures and Algorithms, Data Science I-IV, Statistical Reasoning and Computation in Data Science, Computer Science I-IV, Statistics in R, Calculus I-III, Discrete Math, Matrices, Data Ethics

TECHNICAL SKILLS

Languages & Frameworks: Python, C#, .NET Framework, SQL

AI & Automation: Large Language Models (LLMs), Prompt Engineering, Ollama, n8n, Model Context Protocol (MCP), OCR, Vision-Language Models (VLMs)

Cloud & DevOps: Azure DevOps, Microsoft Azure, Azure SDK, Azure Key Vault, Kubernetes, Docker, Git, CI/CD

Data & Integrations: Snowflake, SQL Server, Redis, Microsoft Graph, REST APIs, Pandas, RapidFuzz

INTERNSHIP EXPERIENCE

Pennsylvania Compensation Rating Bureau (PCRB) - AI Engineer Intern

Summer 2026 | Philadelphia, PA

- Built the AI Gov Weekly Release workflow in **Python**, querying on-prem **Azure DevOps REST APIs/WIQL** for weekly release work items, feeding ticket context to locally hosted **LLMs**, and converting engineering details into concise business-facing release summaries.
- Applied **Prompt Engineering** to produce deterministic local-LLM release notes: a one-sentence weekly preview and 1-3 sentence business-impact summaries per ticket, excluding employee names for consistency.
- Deployed on-premise inference with **Ollama**, **Docker**, **Kubernetes**, and **Redis**, including containerized multi-replica model serving, client-side request distribution, concurrency controls, and internal API access while keeping company data off external AI services.
- Integrated AI workflows with **C#**, **.NET Framework**, **Azure SDK**, **Azure Key Vault**, **Microsoft Graph**, and **Azure DevOps REST APIs/WIQL**, using certificate-based authentication and configuration-driven secret retrieval for secure enterprise automation.
- Developed ticket-lifecycle and software-delivery compliance analysis that detects skipped states, reopens, missing commits/PRs, insufficient reviewer approvals, unresolved review feedback, and business sign-off issues; surfaced findings through **CI/CD** reporting.
- Designed deployment and pipeline analytics comparing production and non-production activity, tracking execution times, run frequency, success/failure trends, rollback evidence, changed-file volume, agent-pool errors, and failure classifications with AI-generated explanations linked to source evidence.
- Implemented **Kubernetes** EndpointSlice/RBAC service discovery and client-side load balancing to distribute concurrent local-model requests across replicas while maintaining per-worker concurrency limits.
- Built **Azure DevOps** pipeline failure diagnostics that classify issues as application, infrastructure/agent, security, deployment/environment, or unknown and provide direct links to supporting logs and engineering evidence.
- Designing an agentic payroll audit and document-processing workflow using **n8n**, **Model Context Protocol (MCP)**, **OCR**, **Vision-Language Models (VLMs)**, and local AI agents to extract, validate, route, reconcile, and review payroll information across internal systems.
- Built interactive **Git** and **Azure DevOps** utilities to search and filter commits and pull requests, link selected development artifacts to work items, and improve traceability across code changes and tickets.
- Automated Weekly Release email delivery by assembling HTML reports with ticket summaries and direct **Azure DevOps** links, then securely distributing them through **Microsoft Graph**; extended the same reporting framework to monthly, quarterly, and yearly engineering reports.
- Designed payroll-workflow privacy and quality controls including PII placeholder redaction/restoration, authentication at each stage, human-in-the-loop approval before publication, and feedback loops that route incorrect outputs back to specialized agents for correction.

Pennsylvania Compensation Rating Bureau (PCRB) - Data Science / Actuarial Research Intern

Summer 2025 | Philadelphia, PA

- Improved address and branch matching accuracy by 4% with scalable **Python** pipelines using **Pandas**, **RapidFuzz**, and regex for standardization and fuzzy matching across millions of records.
- Built traceable validation and anomaly-detection workflows and queried, validated, and transformed large datasets using **Snowflake**, **SQL Server**, and **SQL**.
- Automated anomaly detection on third-party data to flag inconsistencies before production, strengthening proactive data-quality checks across externally sourced records.
- Proposed spatial reconciliation, **REST API** validation, and **Microsoft Azure** automation approaches to improve long-term data quality; several recommendations moved into implementation.

ACTIVITIES

- Ri3D Club:** Built a functional robot in 72 hours for the annual FIRST Robotics challenge.
- Nittany AI Society:** Participate in AI workshops, hackathons, and applied machine learning activities.
- Nittany Cloud Association:** Hands-on exposure to Azure, AWS, and Google Cloud platforms.